The $20 Million Question

"In a public health crisis, every dollar spent on the wrong outreach can cost lives."

How do we predict who will say 'Yes' in advance?

The Mission

- Transform health 2009 H1N1 data from **reactive guesswork → predictive precision**
- Identify key drivers of vaccine uptake
- Target the right communities at the right time
- Measure the population's **Trust Threshold**

Key Insight

- **This is not just a model — it's a strategic decision-making tool**

Prepared and Presented By: Abel Aleu Chol Garang | Data Scientist

The Data Blueprint & Strategic Gap

Decoding Hidden Behavioral Patterns

The Core Conflict

"Vaccine hesitancy is not just personal — it is a barrier to herd immunity."

- When public health relies on guesswork:
- Resources are wasted
- Outreach becomes ineffective

The Dataset: 26,707 individuals | 35 variables

- **Psychological:** Risk perception, trust, opinions
- **Behavioral:** Hygiene habits, preventive actions
- **Demographic:** Income, education, access

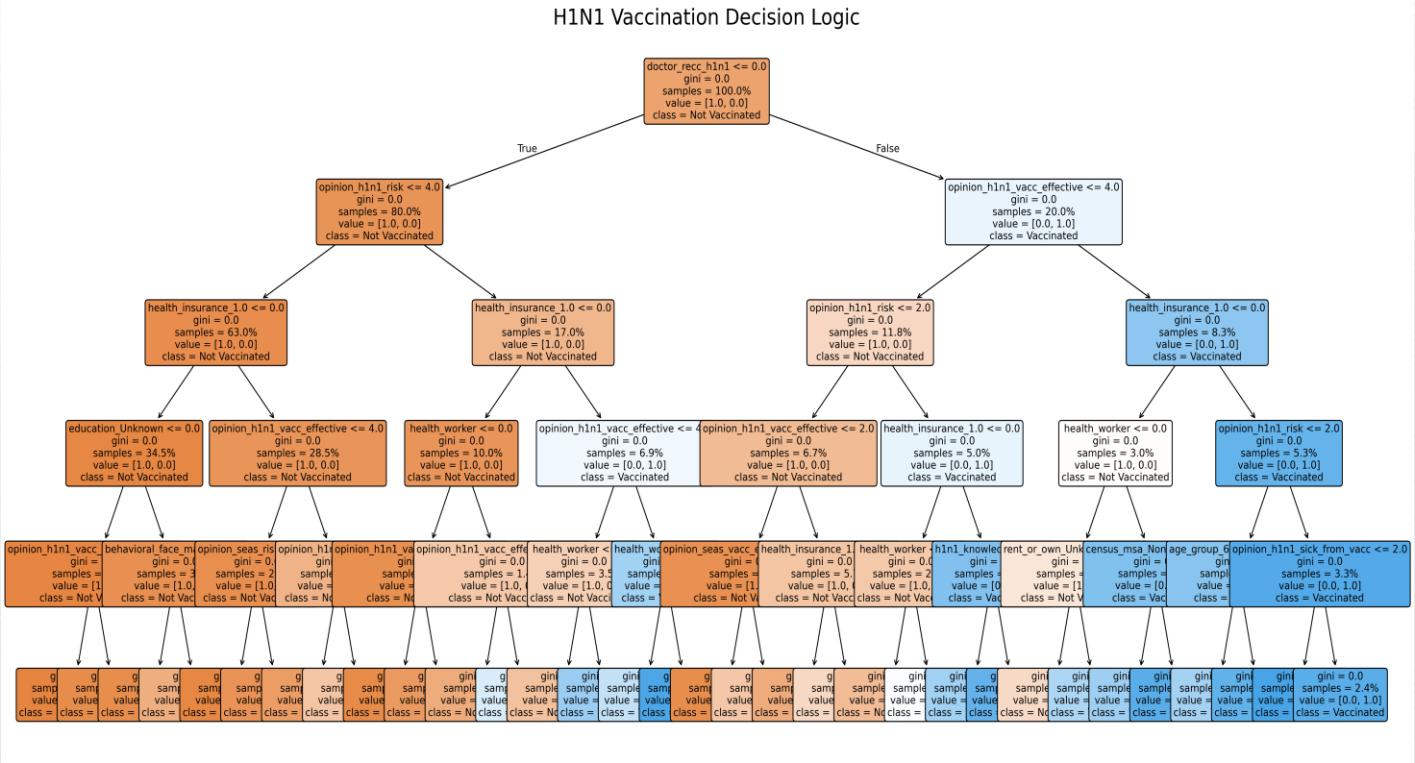
The Challenge:

- **Only 21% vaccinated → a hidden minority**
- Difficult to identify ("needle-in-a-haystack")
- Traditional approaches miss them
- Requires a high-recall model

Key Insight

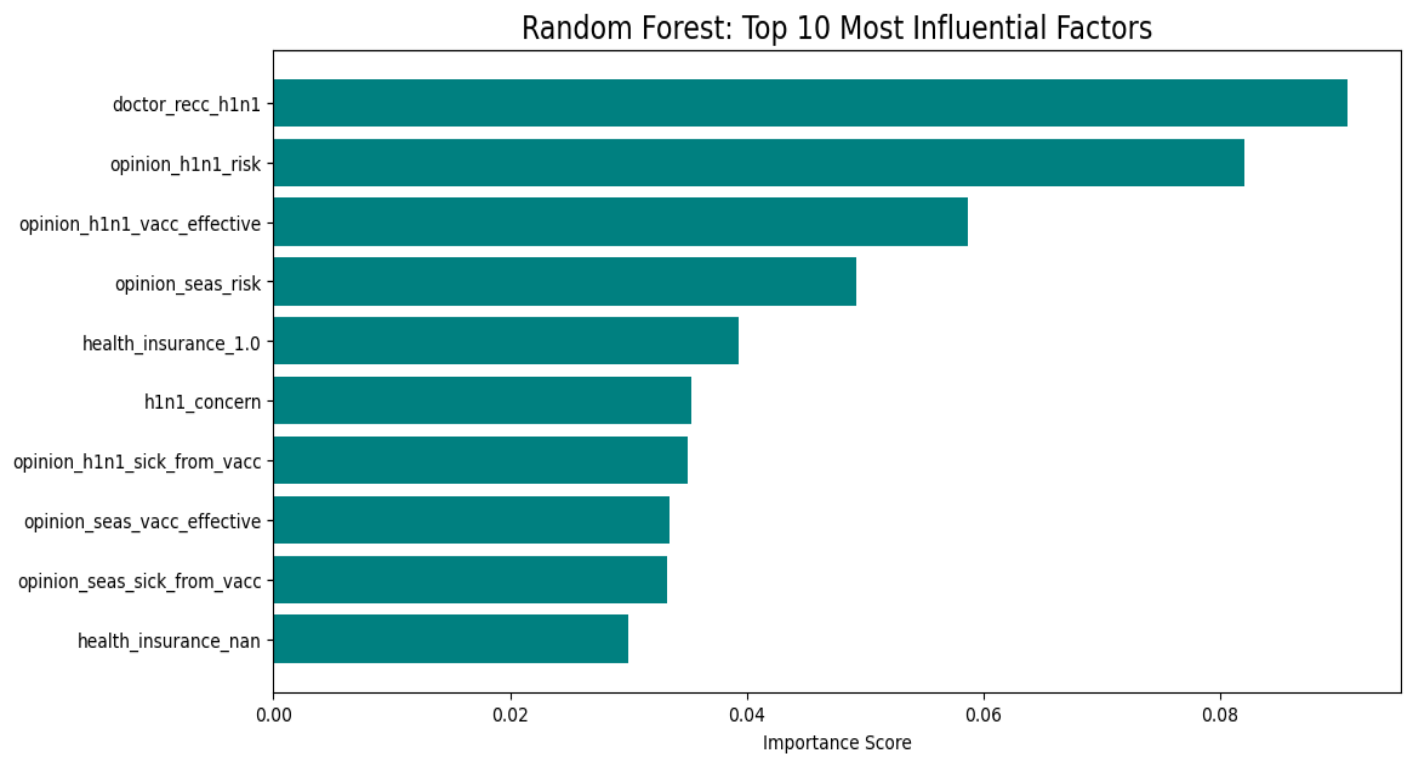
- **From mass marketing → precision outreach**

Decision tree Model



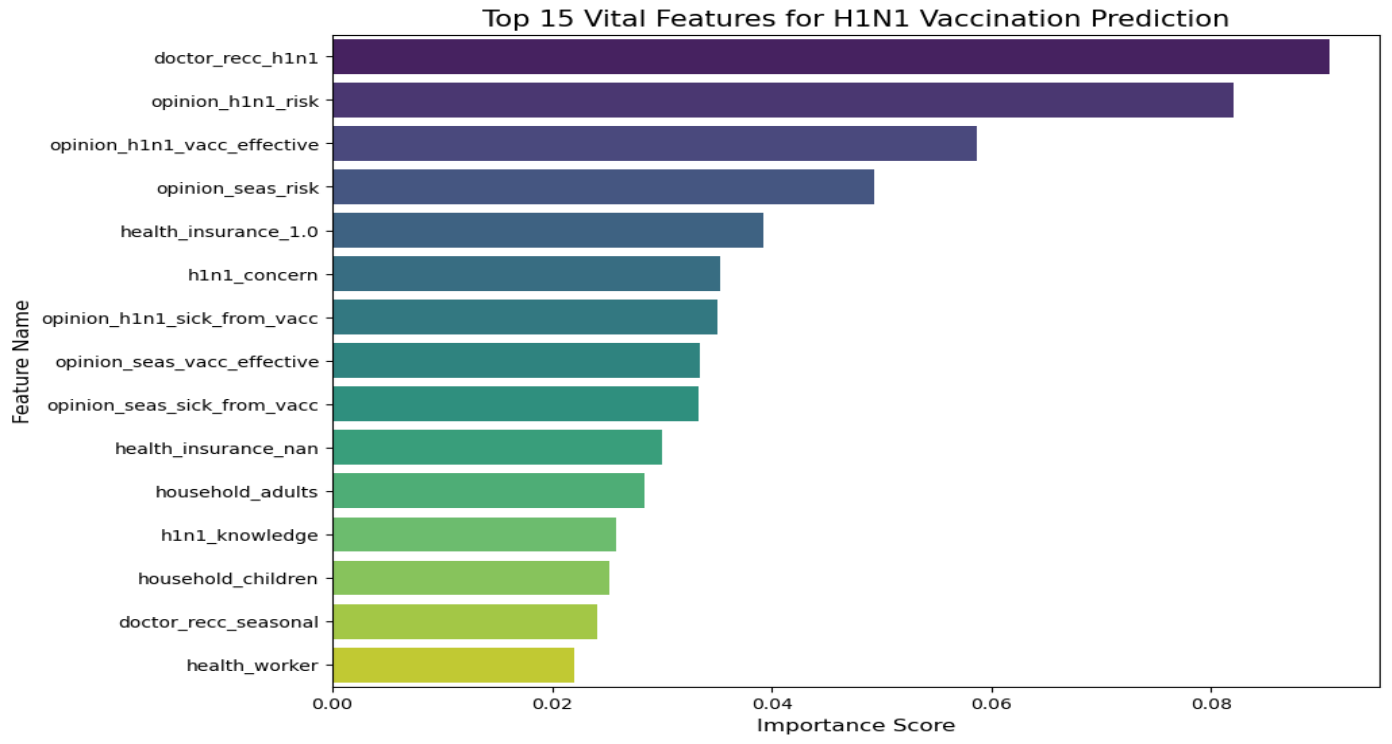
Random Forest Model

Perception vs. Reality



The Hierarchy of Influence

The most influential factors driving an individual's decision to get vaccinated.



The Anatomy of Decision-Making

Trust vs. Habit: What Really Drives Vaccination

The Data Landscape 26,707 individuals | 35 variables

- **Who:** Age, race, income, education
- **Actions:** Hygiene habits, preventive behavior
- **Beliefs:** Risk perception, vaccine confidence
- **Challenge:** 21% vaccinated → class imbalance

The Strategic Discovery

- "Vaccination is not a habit — it is an act of trust."
- Hygiene behaviors → weak predictors
- **Trust in vaccine effectiveness → strongest driver**
- Concern alone does not translate into action

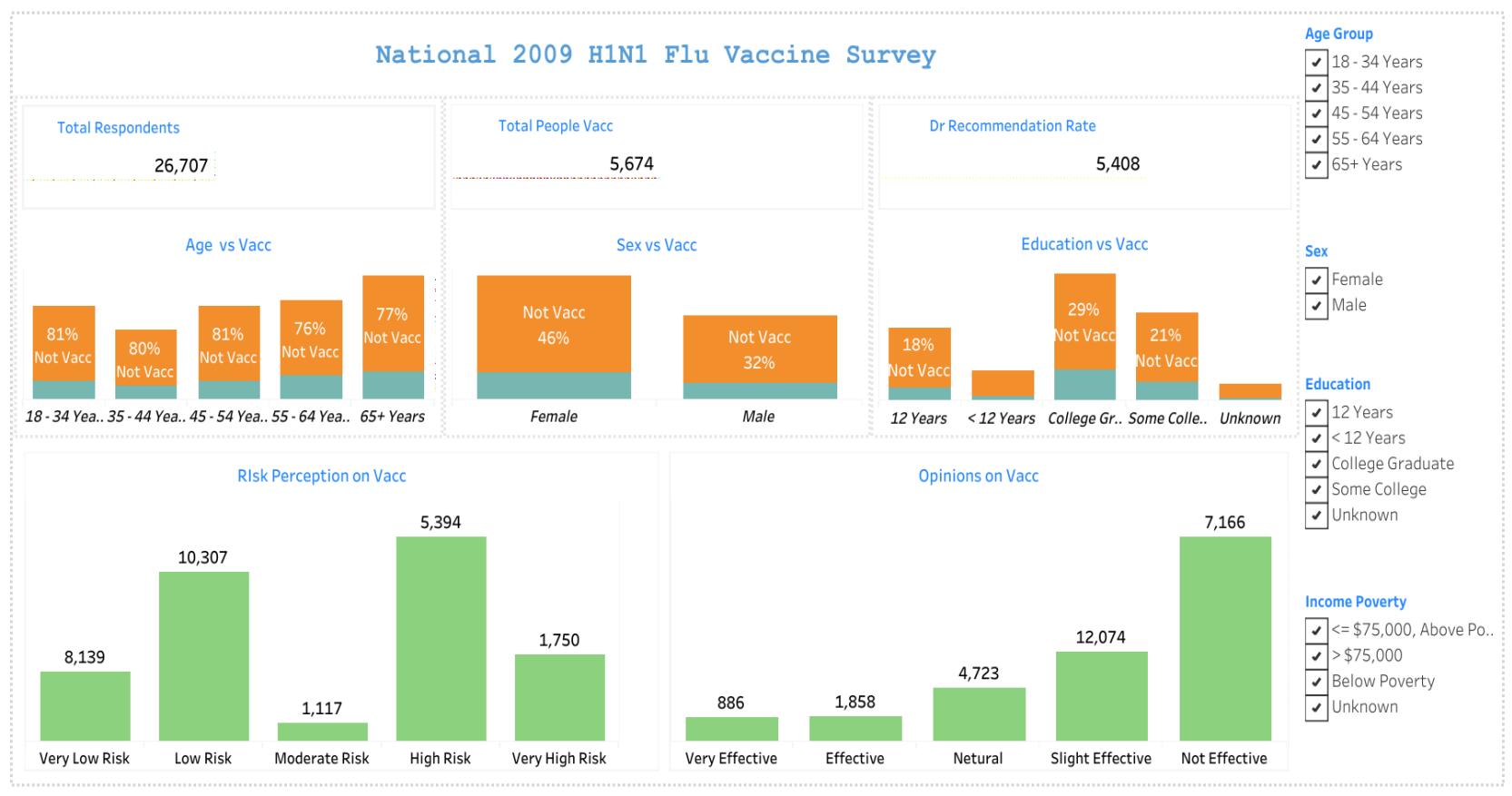
Strategic Shift

- Move from **general awareness → trust-building communication**
- From: "Wash your hands"
- To: **"Why the vaccine works"**

Key Insight

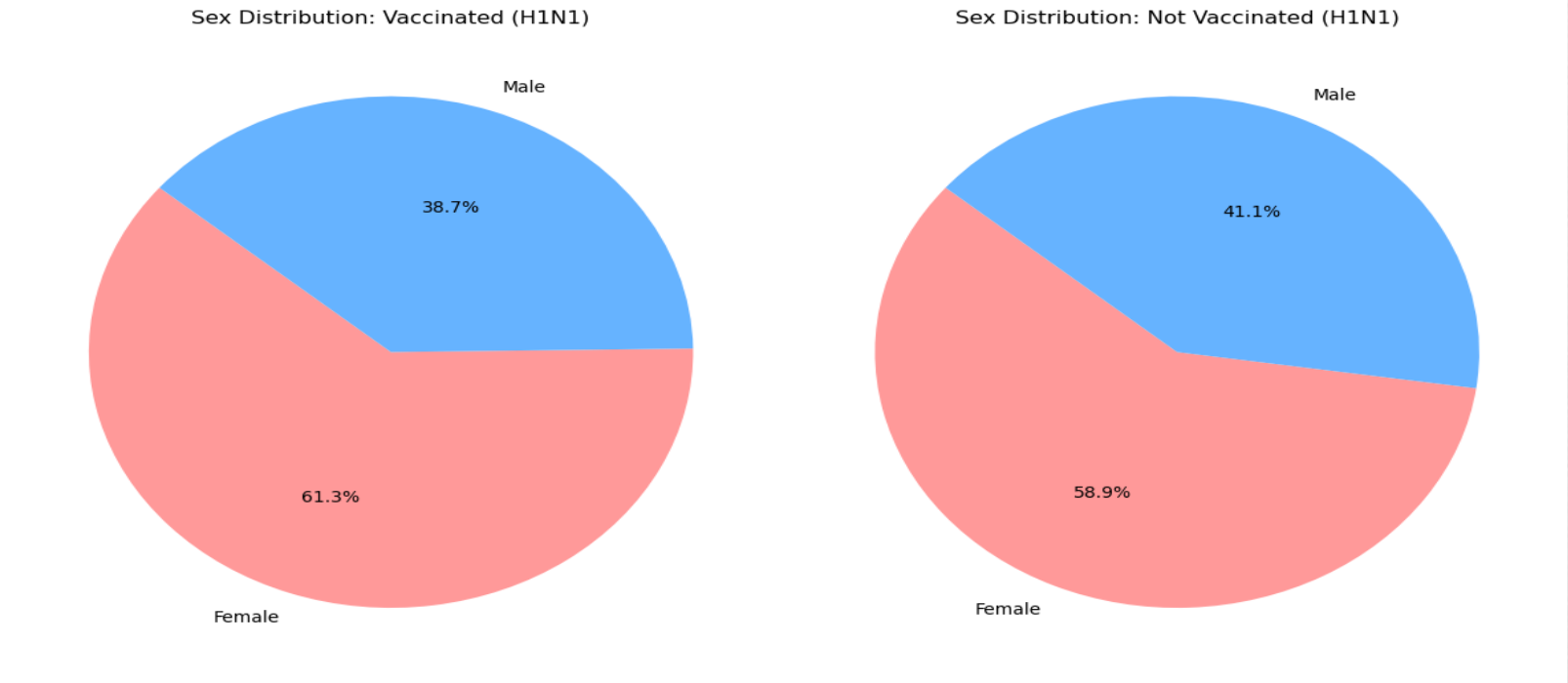
- **Trust, not habit, determines vaccine uptake**

Tableau Representation of the analysis



Vaccinated vs. Unvaccinated

The slices are nearly identical, indicating that **gender was not a primary driver** for H1N1 vaccination allowing us to focus on **behavioral drivers**



Engineering Equity & The Information Gap

From Data Processing to Socioeconomic Insight

Data Preparation

- Imputation: handled missing values
- Encoding: converted categorical variables
- Scaling: normalized numerical features

The Discovery

- **Vaccination gap = Information gap**
- Lower income & education → lower uptake
- Complex messaging limits understanding
- Trust breaks where clarity is missing

Strategic Shift

- **From academic jargon → community-first communication**

Key Insight

- **Access to clear information drives equity**

Escaping the Accuracy Trap

Prioritizing Recall Over Accuracy

The Core Discovery

- Hygiene ≠ vaccination
- **Trust in effectiveness = key driver**

The Problem with Accuracy

- 79% accuracy possible by predicting "No"
- High accuracy can still fail in practice

The Strategic Pivot

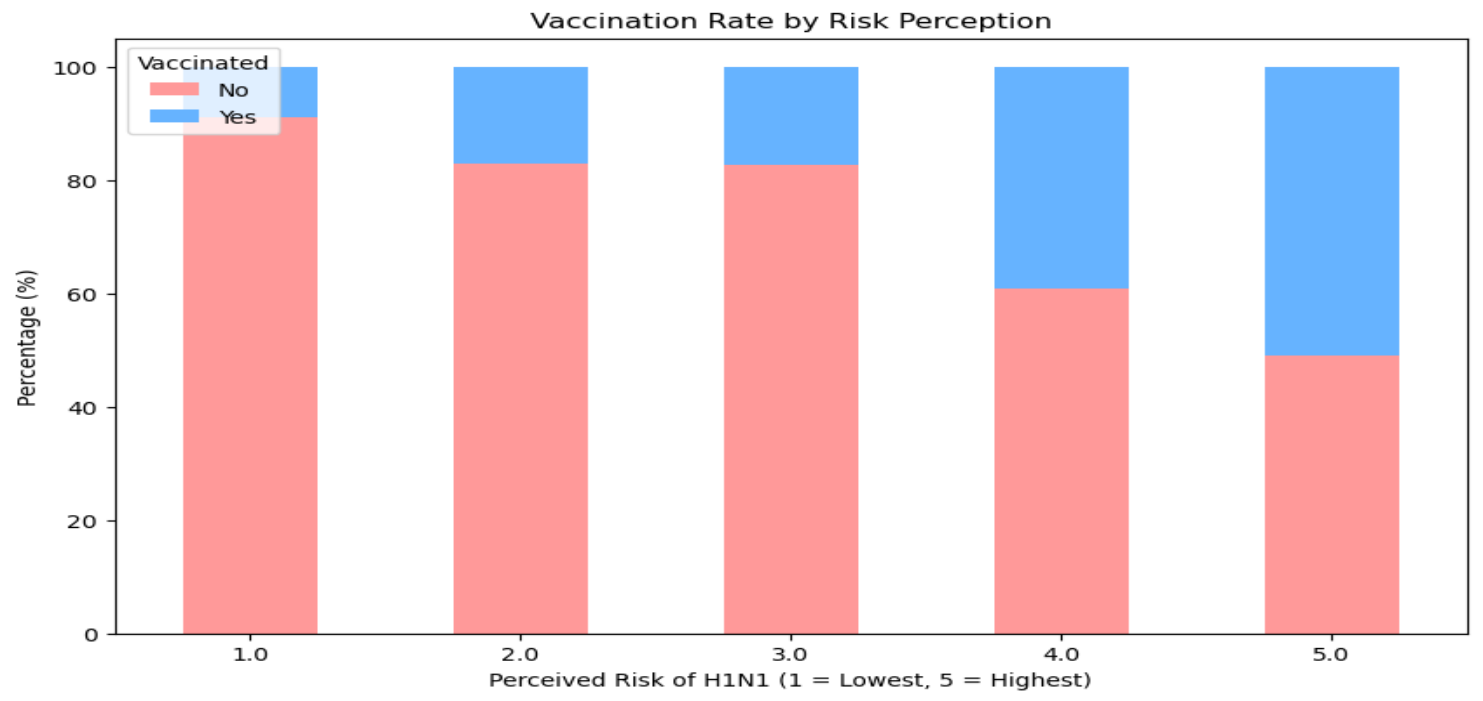
- Focus on **Recall (find potential vaccinees)**
- Minimize missed opportunities

Key Insight

- **Better to reach more than to miss the willing**

Perception of Risk as a Vaccination Drive

The upward trend in the blue segments shows that while few who felt **Low Risk** got vaccinated, nearly half of those perceiving **Highest Risk** took action.



The Strategy of Selection

Choosing Coverage Over Precision

The Gap Identified

- Socioeconomic divide in vaccination
- Underserved groups missed by outreach

Model Comparison

- Random Forest → High accuracy, low coverage
- Logistic Regression → Lower accuracy, **high recall**

The Decision

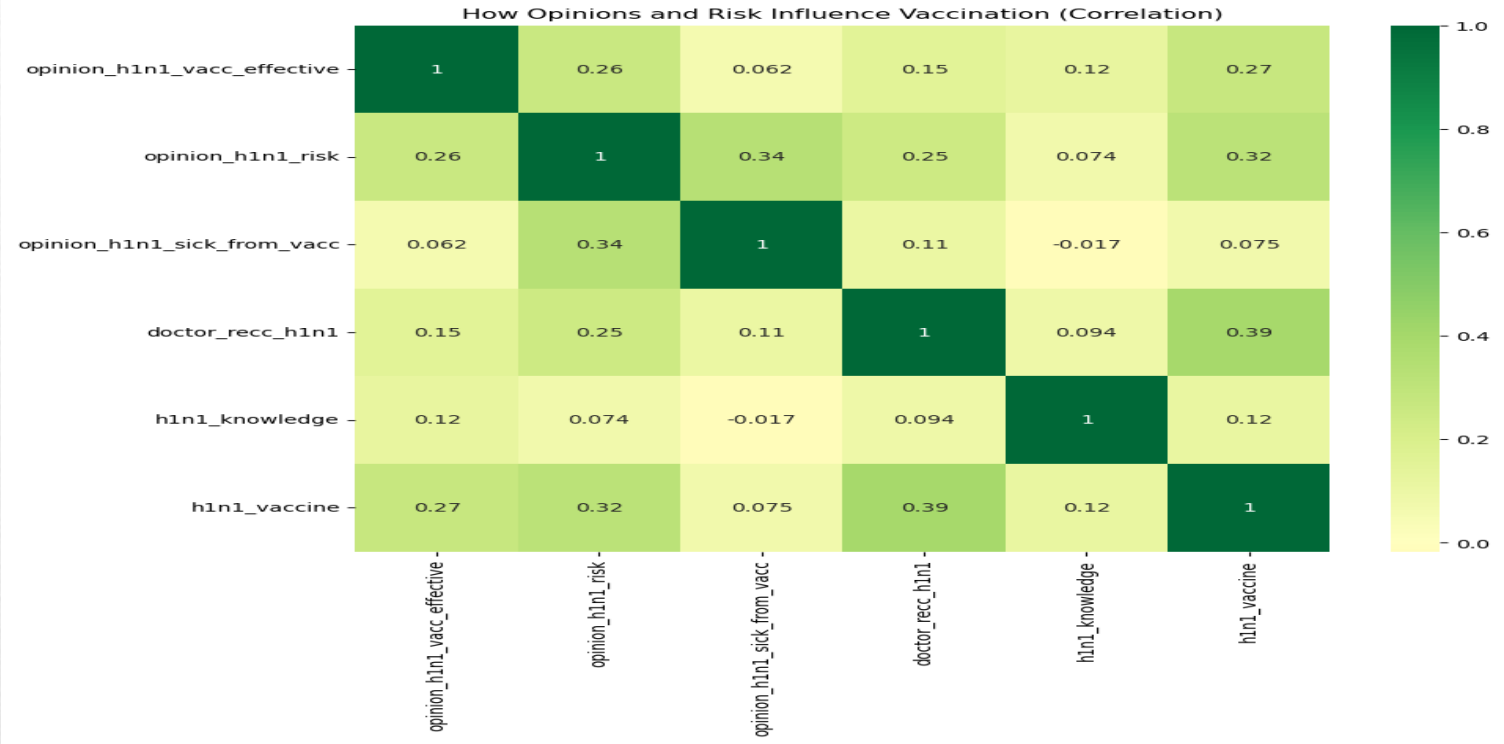
- **Selected: Logistic Regression (Coverage Specialist)**
- Captures more potential vaccinees
- Reduces missed outreach opportunities

Key Insight

- **Coverage saves more lives than precision**

The Interconnected Drivers of Vaccination

Notice the strong positive correlation between **Doctor Recommendation** and **Vaccination Status**. This confirms that clinical intervention isn't just a minor factor.



The Predictive Engine

From Data to Strategic Action

Robust Foundation

- Cleaned, encoded, and scaled data
- Built for real-world generalization

Model Performance

- **ROC-AUC: 0.80+**
- Strong ability to distinguish “Yes” vs “No”

Strategic Optimization

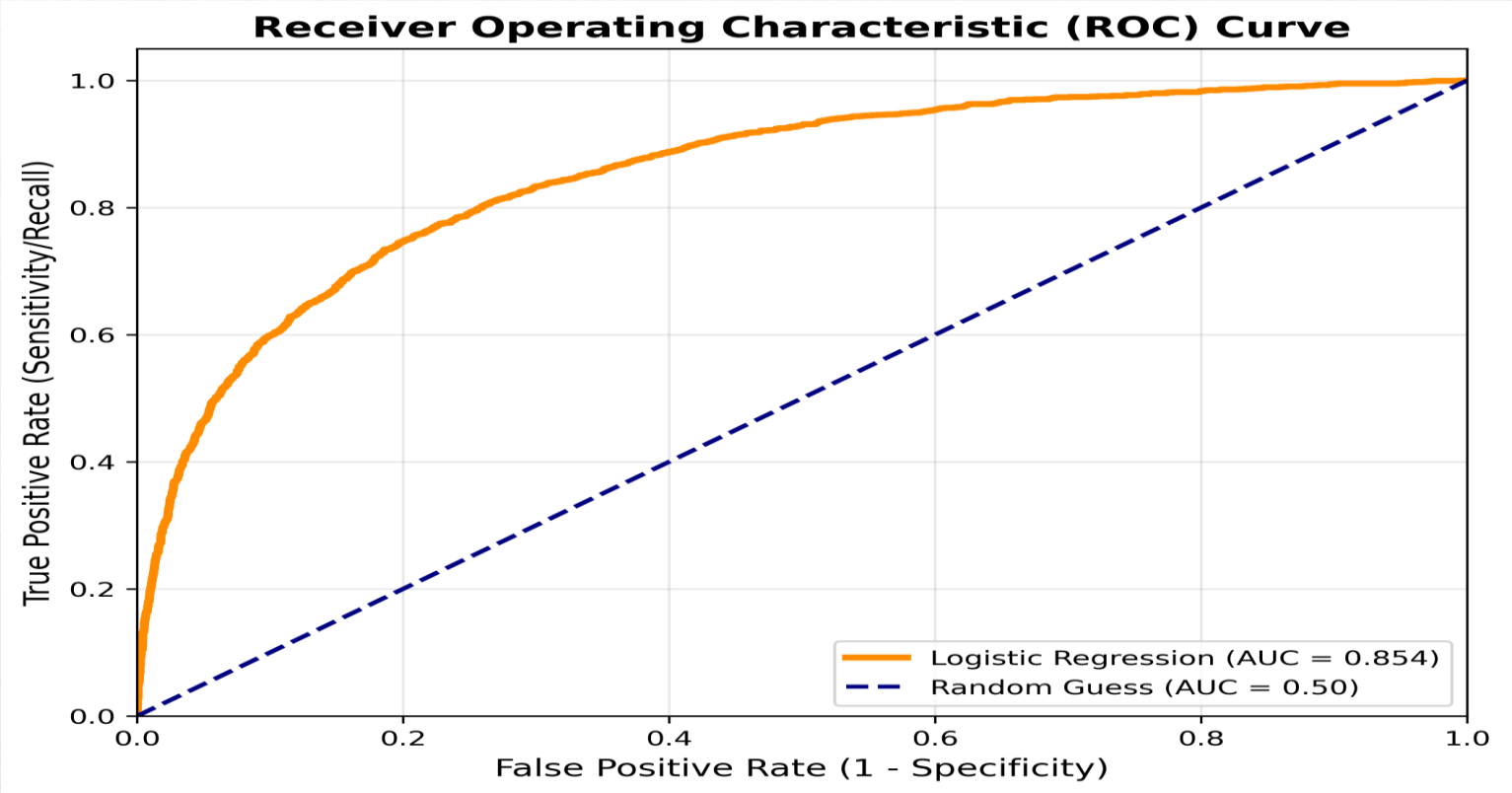
- Prioritized recall over accuracy
- Avoided overfitting

Key Insight

- **A useful model finds people, not just patterns**

Predictive Confidence Index

At **0.85**, the model has an **85% success rate** in telling the 'Yes' group apart from the 'No' group.



The Predictors of Impact

What Actually Drives Vaccination

Model Strength

- High discrimination (ROC-AUC 0.80+)
- Recall-focused → fewer missed vaccinees

Top 3 Drivers

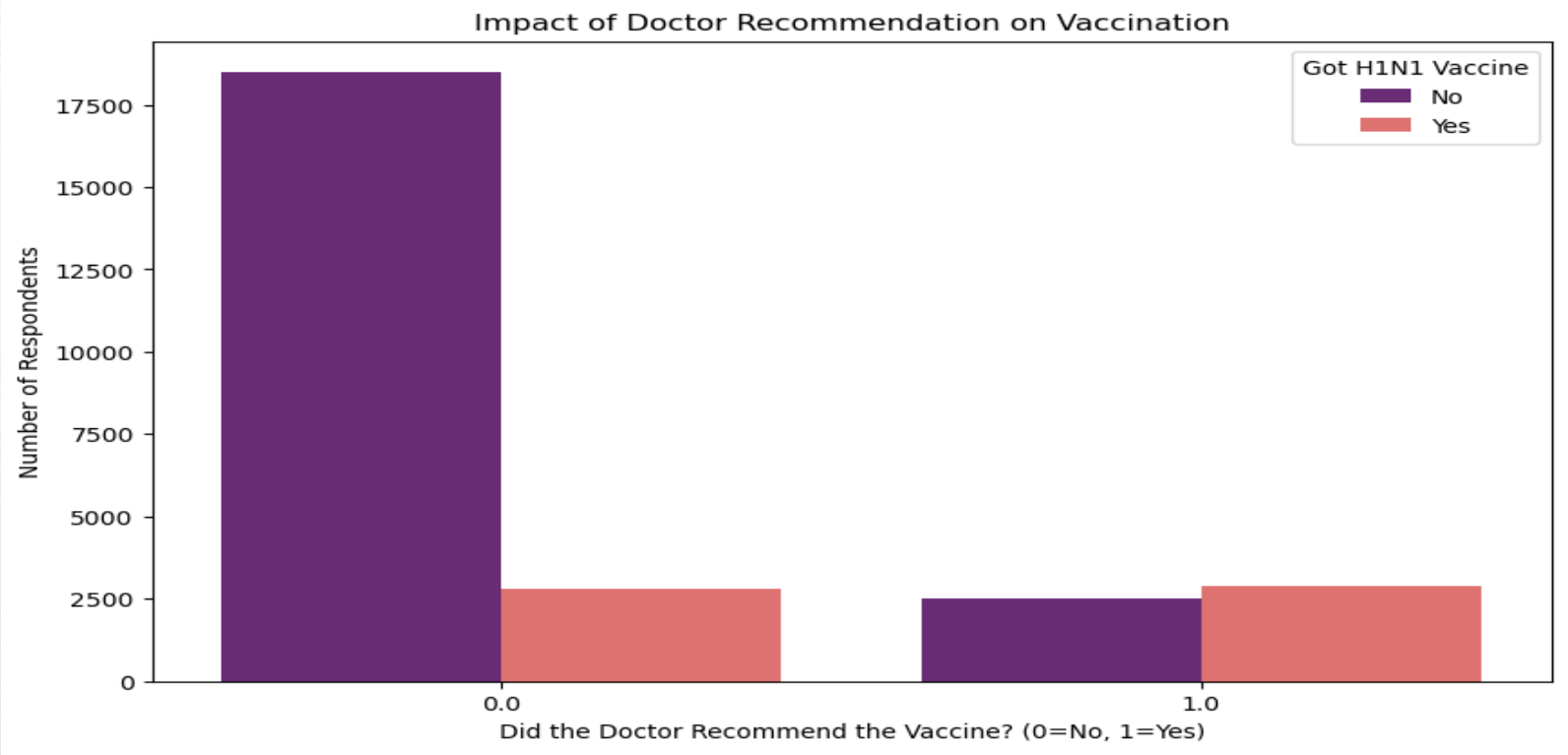
- **Doctor Recommendation** (Strongest)
- **Past Behavior** (Flu shot history)
- **Trust in Effectiveness**

Key Insight

- **Trust + influence + habit = action**

The 'Doctor Effect'

The group on the right (**1.0**) shows a higher vaccination rate despite being smaller. This indicates that a **conversation with a healthcare provider** is one of the most effective ways to increase vaccination uptake.



From Prediction to Protection

A Strategic Action Plan

Key Drivers

- Doctor influence
- Behavioral history
- Trust in vaccine effectiveness

Action Plan

- **Empower doctors** as primary messengers
- **Simplify communication** for underserved groups
- **Target flu-shot individuals first**

Key Insight

- **Strategy turns predictions into impact**

The Future of Public Health Intelligence

Closing the Trust Gap with Data

Final Verdict

- **Vaccination is driven by trust, not just health awareness**
- Model captures ~72% of potential vaccinees
- Enables proactive intervention

Strategic Roadmap

- Equip doctors with communication tools
- Eliminate jargon in outreach
- Track trust & information sources

Closing Statement

- "Data-driven trust is our strongest defense."

Dialogue

Bridging Data & Strategy

Quick Summary

- **Model:** High-recall Logistic Regression maximizes community coverage
- **Insight:** Trust outweighs habits in driving vaccination.
- **Strategy:** Clear, targeted communication to close information Gap.

Discussion

welcome your technical inquiries, strategic critiques, or thoughts on how we can better map the 'Trust Threshold' for the next global health challenge.

For Questions, Inbox me at garangabel262@gmail.com. I will get to you within 24 hours.